Asymptotic analysis of solutions for boundary problems for the game-theoretic p-Laplacian

Diego Berti

University of Florence, Italy

Boundary value problems for the game-theoretic p-Laplacian

The **game-theoretic p-Laplacian** is defined formally by

$$\Delta_p^G u := \frac{1}{p}|\nabla u|^{2-p}\operatorname{div}\{|\nabla u|^{p-2}\nabla u\},$$

for $p \in [1,\infty]$, where for Δ_∞^G we intend the limit of Δ_p^G as $p \to \infty$. Let Ω be a bounded domain in $\mathbb{R}^N$, with $N \geq 2$ and let Γ be its boundary. We have considered two problems for Δ_p^G.

Initial-boundary value problem for the evolution game-theoretic p-Laplacian. We want to study the behavior for short times of $u = u(x,t)$, the viscosity solution of

$$\begin{cases} u_t - \Delta_p^G u = 0 & \text{in } \Omega \times (0,+\infty), \\ u = 1 & \text{on } \Gamma \times (0,\infty), \\ u = 0 & \text{on } \Omega \times \{0\}. \end{cases} \qquad (1)$$

Boundary value problem for the resolvent equation. Given $\varepsilon > 0$, we want to derive asymptotics, for small values of the parameter, of the viscosity solution of $u^\varepsilon : \Omega \to \mathbb{R}$, of the following elliptic problem:

$$\begin{cases} \varepsilon^{-2}u^\varepsilon - \Delta_p^G u^\varepsilon = 0 & \text{in } \Omega, \\ u = 1 & \text{on } \Gamma. \end{cases} \qquad (2)$$

Remarks

Remark 1. Δ_p^G is related to the well-known p-Laplacian, $\Delta_p u = \operatorname{div}\{|\nabla u|^{p-2}\nabla u\}$, with two fundamental differences: Δ_p^G is **one-homogeneous** and is **not in divergence form**.

Remark 2. Since Δ_p^G is not defined at spatial critical points of u, we define the equations in (1) and (2) in terms of the theory of **viscosity solutions**.

Remark 3. A special mention is needed about the **radial case**, when Δ_p^G acts as a linear operator. If Ω is a ball or the complement of a ball, solutions of (1) and (2), which are radial symmetric functions, are given in terms of *Bessel functions* and *modified Bessel functions* and precise estimates on asymptotics are established.

Varadhan-type formulas

Setting. Given $\omega : (0,\infty) \to (0,\infty)$, a continuous function with $\omega(0^+) = 0$; we say that $\Omega \in C^{0,\omega}$ if Γ is locally a graph of a continuous function with modulus of continuity controlled by ω. In Figure 1, $x \in \Omega$ and $y_x \in \Gamma$ is the point realizing $|x - y_x| = d_\Gamma(x)$, where, d_Γ means the distance function from Γ.

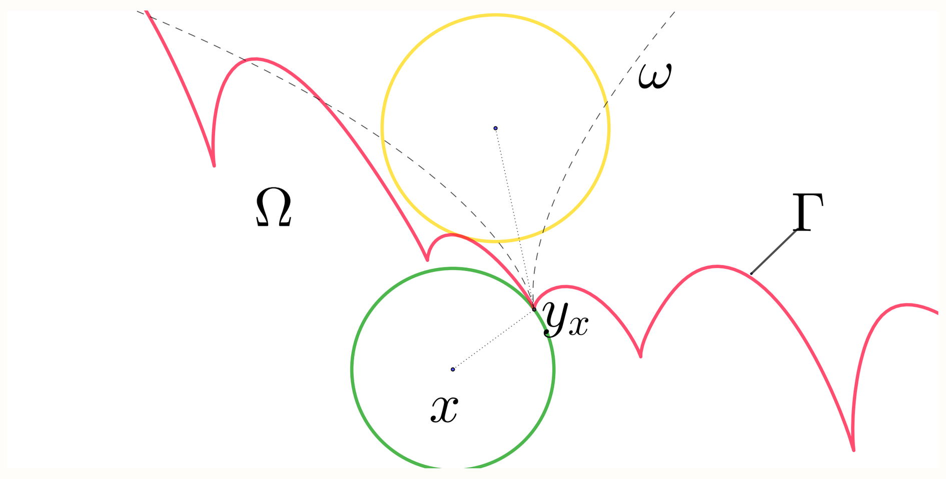

Figure 1: A picture representing a $C^{0,\omega}$ domain.

Let $p \in (1,\infty]$, $p' = p/(p-1)$ and set

$$\psi_\omega(\sigma) = \inf\{\sqrt{s^2 - (\omega(s) - \sigma)^2} : s \geq 0\},$$

for $\sigma > 0$, then we prove the following two theorems.

Theorem ([BM1]) *Let $\Omega \in C^{0,\omega}$ and u be the (viscosity) solution of (1). Then, as $t \to 0^+$,*

$$|4t\log u(x,t) + p' d_\Gamma(x)^2| = O(t\log\psi_\omega(t)).$$

Remark 4. The previous theorem is obtained essentially taking advantage on the geometrical scheme in Figure 1, **by approximating $u(x,t)$ with radial sub-solutions and super-solutions of (1).**

Remark 5. If Ω satisfies sufficient regularity assumptions (for example, if Γ is **uniformly Hölder continuous**), we obtain the **uniform convergence on $\overline{\Omega}$**, with rate $O(t\log t)$.

Theorem ([BM2]) *Let $\Omega \in C^0$ and u^ε be the solution of (2) and denote by $\xi(\varepsilon,x) = |\varepsilon\log u^\varepsilon(x) + \sqrt{p'}d_\Gamma(x)|$, for $x \in \Omega$ and $\varepsilon > 0$. Then, as $\varepsilon \to 0^+$,*

$$\xi(\varepsilon,x) = \begin{cases} O(\varepsilon) & \text{if } p = \infty \\ O(\varepsilon\log\varepsilon) & \text{if } p > N. \end{cases}$$

Assume that $\Omega \in C^{0,\omega}$. Then, as $\varepsilon \to 0^+$,

$$\xi(\varepsilon,x) = \begin{cases} O(\varepsilon\log|\log\psi_\omega(\varepsilon)|) & \text{if } p = N \\ O(\varepsilon\log\psi_\omega(\varepsilon)) & \text{if } 1 < p < N. \end{cases}$$

Remark 6. Same observations as in Remarks 4 and 5 can be done. In this case, a more precise qualitative dependence on the sign of $N - p$ is established. This fact is due to the **more explicit knowledge of radial solutions.**

Asymptotics for q-means

Let $\Omega \in C^2$, $x \in \Omega$ and $R = d_\Gamma(x)$. In [BM1, BM2] we also established asymptotic formulas for the so-called q-*means* on a ball touching the boundary. The **q-mean** of u on $B_R(x)$, denoted by $\mu_{q,t}(x)$, is the minimizer of $\|u(\cdot,t) - \lambda\|_{L^q(B_R(x))}$, $\lambda \in \mathbb{R}$.

Theorem ([BM1]) *Let $q \in (1,\infty)$ and u be the solution of (1). Then:*

$$\lim_{t\to 0^+} (R^2/t)^{\frac{N+1}{4(q-1)}}\mu_{q,t}(x) = C_{N,q}\,p'^{-\frac{N+1}{4(q-1)}}\Pi_\Gamma(y_x)^{-\frac{1}{2(q-1)}},$$

where $C_{N,q}$ is a positive constant.

Here,

$$\Pi_\Gamma(y) := \prod_{j=1}^{N-1}[1 - R\kappa_j(y)],$$

where $\kappa_1,\dots,\kappa_{N-1}$ are the principal curvatures of Γ. **Remark 7.** A similar result for the elliptic problem is proved in [BM2]:

$$\lim_{\varepsilon\to 0^+} (R/\varepsilon)^{\frac{N+1}{2(q-1)}}\mu_{q,\varepsilon}(x) = \tilde{C}_{N,q}\,p'^{-\frac{N+1}{4(q-1)}}\Pi_\Gamma(y_x)^{-\frac{1}{2(q-1)}},$$

where $\tilde{C}_{N,q}$ is a positive constant.

Symmetry results

Here, an application of previous formulas is reported.

Theorem ([BM3]) *Let Ω be a C^2 domain, with bounded boundary Γ. Assume that there exists a hypersurface $\Sigma \subset \Omega$ of class C^2, such that $u(x,t)$ is constant on Σ, at each time $t > 0$. Then, Ω must be radially symmetric.*

Remark 8. In the previous theorem, we can weaken the regularity assumptions on Σ, by considering the work of Maz'ya et al. [ABMMZ]: the statement still holds if Σ satisfies a proper interior **pseudo-ball condition**, in the sense of [ABMMZ, Theorem 4.4].

References

[BM1] Berti, Magnanini, J. M. P. A., (to appear) 2018.

[BM2] Berti, Magnanini, A. A., 2018.

[BM3] Berti, Magnanini, in preparation.

[ABMMZ] Alvarado, Brigham, Maz'ya, Mitrea and Ziadé, J. M. S., 2011.